



Anil Seth

Is Ubuntu's Package Management The Right Thing to Do?

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There have been an impressive number of attempts to build a better mousetrap for package management software. Some believe that the absence of an easier, intuitive package manager is a major reason preventing wider adoption of the Linux desktop. I might not agree with this but, with the release of the new version of 'Ubuntu Software Centre' on Ubuntu 10.04, an interesting question to consider is whether those at Ubuntu are doing the right thing?

I have very fond recollections of the philosophy courses I studied—they taught me that the answer to the above question is not as easy as it may seem. (As an aside, to know whether you know the right thing to do, you may wish to see the first lecture at <http://www.justiceharvard.org/>)

When you open Ubuntu Software Centre, you are shown a reasonable grouping of applications. We will explore the multimedia options. In fact, we will confine ourselves to the video players only. Let's pretend to be a new user who has no idea about the available video players, and is determined to find the one he likes best. We will then use tools like *dpkg-query* and Synaptic to understand what is happening behind the scenes.

Gnome-mplayer seems nice so we install it. As expected, the package manager does its work, and does not display any more messages or inform us of the various dependencies that will be installed. Using *dpkg-query*, we confirm that it has indeed installed over 20 packages. We now remove this package before trying another. Typically, an excellent utility like Synaptic will fail us: it will uninstall *gnome-mplayer*, but not the other 20-odd packages that were installed to satisfy dependencies. In contrast, Ubuntu Software Centre quietly removes them all.

We try *smplayer*. Installation and removal work equally well and quietly, so let us try a somewhat more complex interaction. We install *gnome-mplayer*, and then install *smplayer*. Now, we uninstall *gnome-mplayer*. The behaviour is reasonable—the common dependencies are not removed. *Smplayer* continues to be functional.

We now remove *smplayer*, and it does not do the right thing. The common dependencies remain. Reinstalling *gnome-mplayer* and removing it, cleans up the packages that had been left behind!

Our next stop is a player I had not tried before—*kmplayer*. As expected, it needed a lot of extra packages because of the KDE dependencies. Surprisingly, the one package not included as a dependency was *mplayer*! Thus, *kmplayer* could not play any file. Worse, it did nothing, and did it very quietly, ignoring all our attempts to show some video.

However, since we are not discussing *kmplayer*, and Kubuntu is the right distribution for KDE applications, we will ignore this anomaly. Incidentally, *smplayer* uses the QT libraries, but not the

KDE libraries. We go ahead and uninstall *kmplayer*. Only one package is removed; all the KDE packages remain!

This was indeed a major disappointment. Efforts to clean up the KDE applications using only the Ubuntu Software Centre resulted in 32 packages being broken as per Synaptic. Any effort to install or remove a package results in Ubuntu Software Centre wanting us to run *apt-get install -f*. We, instead use *apt-get autoremove -f* to clean up the broken packages.

By now, it must be clear that my pain point with package management has been the clutter left behind when one is trying different packages. It is not because of disk space constraints, but because of the wasted time on the updates of unneeded packages. I avoid reinstalling an OS, so the upgrades and updates cover a number of distribution versions. Over a period of time, some updates fail because of the unneeded packages remaining on the system. Aesthetically, just the idea of needless clutter lying on my system is very unappealing.

As a final test, we will install *mplayer* using Synaptic. Synaptic shows it in the list of manually installed packages. We now install *gnome-mplayer* using Ubuntu Software Centre. Verify that it is working, and uninstall it. Does it do the right thing by leaving *mplayer* installed? After all, it was already installed prior to the installation of *gnome-mplayer*. We find that it does do the right thing.

The magic, though imperfect at present, appears to lie in using *apt-get's autoremove* option. So, overall, the progress in the management of packages is very good. Hopefully, the inconsistencies in the *autoremove* capabilities of *apt* will be ironed out. These are probably in the packaging configurations.

The display and grouping of applications relies on the desktop files in */usr/share/app-install*. The system group, though, shows all applications in the 'system' category. I would guess that this category includes all the packages that do not have a desktop file associated with them.

No one complains about the hundred thousand applications available for the iPhone; hence, the remarkable range of applications available for Linux is a feature, not a liability. We just need a nice short description of each package, with a nice keyword search. Ubuntu Software Centre is definitely headed in the right direction. 

About Anil Seth

The author works as a consultant. Prior to consulting, Anil was a professor at Padre Conceicao College of Engineering (PCEE) in Goa, managed IT and imaging solutions for Phil Corporation Limited (Goa), supported domestic customers for Tata Burroughs/ TIL, and was a researcher with IITK and the Indian Institute of Geomagnetism (Mumbai). You can find him online at <http://sethanil.com> and reach him via e-mail at anil@sethanil.com.